

DreamBooth finetuning for pose conditioning T2I

DreamBooth

Given as input just a few images (4-5) of a subject, we fine-tune a pretrained text-to-image model such that it learns to bind a unique identifier with that specific subject.

For the model T2I, as imagen is not open-source, so we will use Stable Diffusion as our baseline for further experiment (to be exact: `runwayml/stable-diffusion-v1-5`)

[task 1]

Reading papers

DDPM (a revision before real task)

```
x0 ~ p_data
t ~ Uniform(0, T)
ε ~ N(0, I)

x_t = q_sample(x0, t, ε)
ε̂ = UNet(x_t, t)

loss = ||ε̂ - ε||²

opt.zero_grad()
loss.backward()
opt.step()
```

Here is just pseudo code for what actually happen in the training code of base DDPM

LDM (main archi for Stable Diffusion)

```
x, y ~ p_data
z = E(x)                # image encoder
c =  $\tau(y)$             # condition encoder (text, layout, etc.)

t ~ Uniform(0, T)
 $\epsilon \sim N(0, I)$ 

z_t = q_sample(z, t,  $\epsilon$ )
 $\epsilon^\wedge = \text{UNet}(z_t, t, c)$     # UNet with conditioning

loss = || $\epsilon^\wedge - \epsilon$ ||2

opt.zero_grad()
loss.backward()
opt.step()
```

Archi for LDM (conditioning T2I with latent space)

DreamBooth

Fine tuning a few samples (4-5) of a subject (a specific human, animal, ...) with 2 tricks:

- a [V] dog
- Class-Specific Prior Preservation Loss

=> make the model learn that [V] is the subject, but still not loss the ability to work on normal prompt

```
x_i ~ instance_images      # "a [V] dog"
x_c ~ class_images         # "a dog"

x = concat(x_i, x_c)
p = concat(p_i, p_c)

z0 = VAE.encode(x)
c = TextEncoder(p)

t ~ Uniform(0, T)
ε ~ N(0, I)

z_t = add_noise(z0, ε, t)
pred = UNet(z_t, t, c)

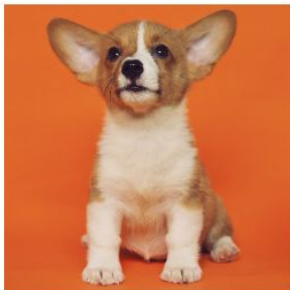
target = ε

pred_i, pred_c = split(pred)
tgt_i, tgt_c = split(target)

loss = ||pred_i - tgt_i||2 + λ ||pred_c - tgt_c||2

opt.zero_grad()
loss.backward()
opt.step()
```

First task: DreamBooth Reproduce

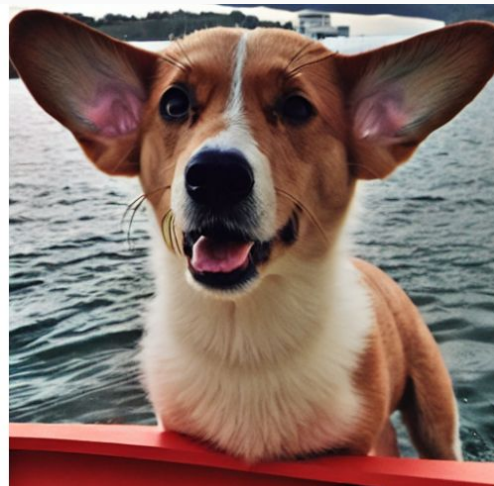


Subject

(from the Dreambooth dataset)



a photo of sks dog on a boat



a photo of sks dog on a boat

Not too prompt fidelity but the model can still learn the subject

First task: code, and how to train the model.

Here is my github code [BFCmath/DreamBooth - dreambooth](https://github.com/BFCmath/DreamBooth-dreambooth)

Second task: Extend Dreambooth

Problem Format:

Given a pretrained diffusion model

3-5 images of subjects

A prompt describe the image want to be generated

A pose

-> Finetuned so that the model can learn - follow prompt, understand the subject (human) and have the input pose

Here is my github code [BFCmath/DreamBooth - controlnet](https://github.com/BFCmath/DreamBooth-controlnet)

Controlnet

Basically the same as standard Stable diffusion, except we clone the a trainable copy of encoder (*while the org encoder is frozen*) and fed the cond_encoder (pose embedding) to create a pose signal -> add to decoder through skip connection

```
z0 = encode_vae(image)
ct = text_encoder(prompt)
cf = cond_encoder(ci)

t = randint(0, T)
eps = normal_like(z0)
z_t = add_noise(z0, eps, t)

h = z_t
ctrl_skips = []

for enc_block, ctrl in zip(sd_enc_blocks_locked, ctrl_enc_blocks_trainable):
    h = enc_block(h, t, ct)
    _, delta = ctrl_block(h, cf, t, ct)
    ctrl_skips.append(delta)

h = sd_middle_locked(h, t, ct)
_, delta_mid = ctrl_middle_block(h, cf, t, ct)
h = h + delta_mid

for dec_block, delta in zip(sd_dec_blocks_locked, reversed(ctrl_skips)):
    h = h + delta
    h = dec_block(h, t, ct)

eps_hat = final_conv(h)

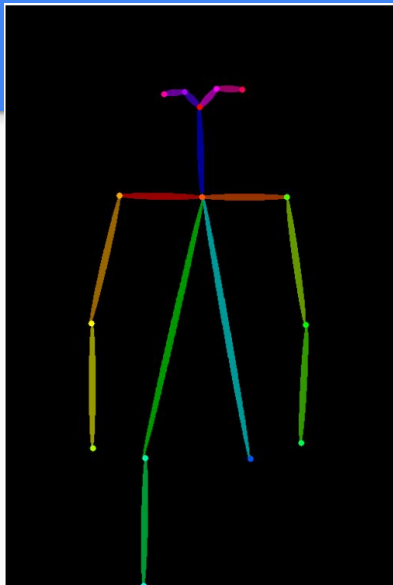
loss = mse(eps_hat, eps)
update(ControlNet_parameters_only)
```



Controlnet

Why controlnet?

- A popular, consistent and modern baseline
- Work well with stable diffusion
- I also check out other paper, check in **Ablation**



a photo of person wearing
jacket, high quality



Controlnet + Dreambooth

So here, we will fine tuning controlnet using very few sample using Dreambooth method so that the controlnet can learn the Subject (Identity)

Thorough task 2, my main task is to find a way to effectively tuning this (as overfitting and underfitting is much more likely to happen in this second task compared to task 1 - which I will prove along the way)

Controlnet + Dreambooth

Code:

[DreamBooth/controlnet/dreambooth_controlnet.py](#) at
[main · BFCmath/DreamBooth](#) (deprecated)
[DreamBooth/controlnet/dreambooth_unet_controlnet.py](#) at [main · BFCmath/DreamBooth](#) (main)

```
x, c_map, y ~ p_instance
x_pr, c_map, y_pr ~ p_class

z $\theta$  = E(x); z $\theta$ _pr = E(x_pr)
c =  $\tau$ (y); c_pr =  $\tau$ (y_pr)
```

(which means I use the pose from
train image to generate prior)

```
t ~ Uniform( $\theta$ , T)
```

```
 $\epsilon$ ,  $\epsilon$ _pr ~ N( $\theta$ , I)
```

```
z_t = q_sample(z $\theta$ , t,  $\epsilon$ )
```

```
z_t_pr = q_sample(z $\theta$ _pr, t,  $\epsilon$ _pr)
```

```
with no_grad():
```

```
    res = ControlNet(z_t, t, c, c_map)
```

```
    res_pr = ControlNet(z_t_pr, t, c_pr, c_map)
```

```
 $\epsilon^{\wedge}$  = UNet(z_t, t, c, res)
```

```
 $\epsilon^{\wedge}$ _pr = UNet(z_t_pr, t, c_pr, res_pr)
```

```
loss_id = || $\epsilon^{\wedge}$  -  $\epsilon$ ||2
```

```
loss_prior = || $\epsilon^{\wedge}$ _pr -  $\epsilon$ _pr||2
```

```
loss = loss_id +  $\lambda$  * loss_prior
```

```
opt.zero_grad()
```

```
loss.backward()
```

```
opt.step()
```

Controlnet + Dreambooth

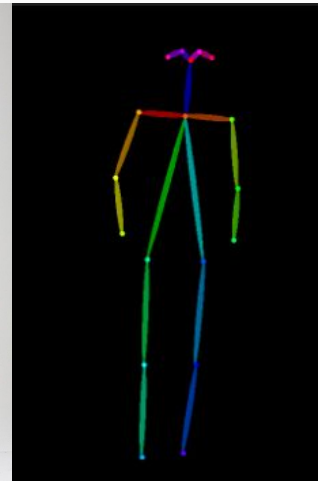
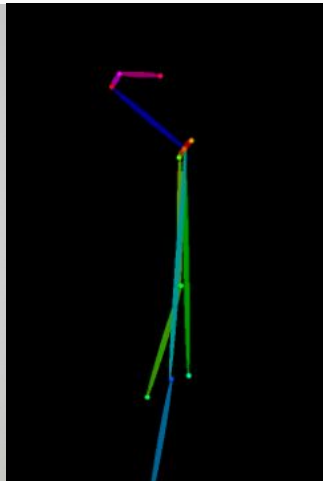
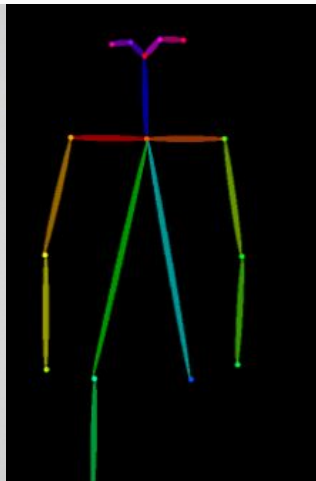


Data: carefully select from

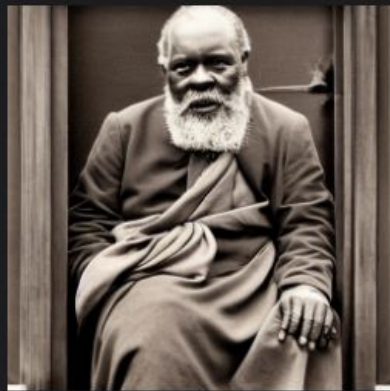
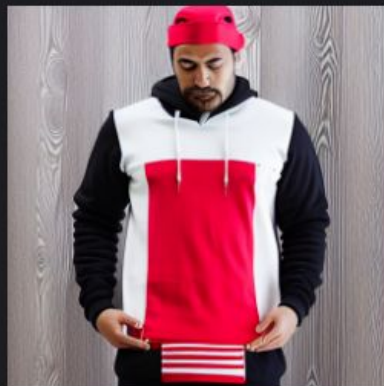
<https://www.kaggle.com/datasets/dreamiiiit/deepfashion-multimodal>

Pose: extracted from dataset use:

l1lyasviel/control_v11p_sd15_openpose
(pose extract model)



Underfit case



Train controlnet
Freeze unet

Train Unet, freeze
controlnet but weak
lora config and lr

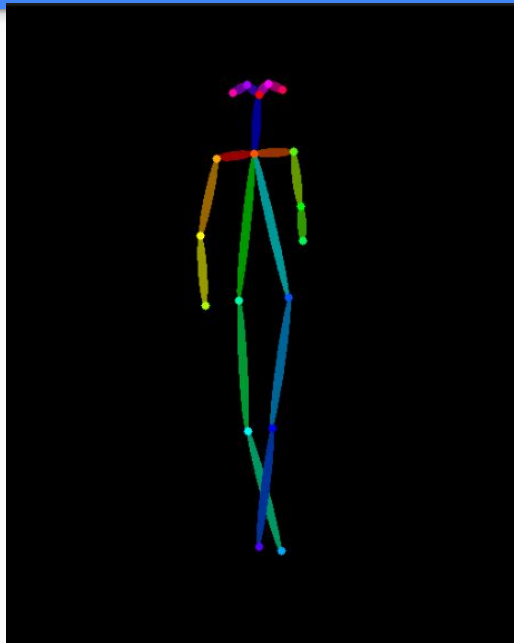
Train text encoder
but weak config

Underfit case

After these, by default I will use:

- Train unet, freeze controlnet (or lightly train) -> reasoning: controlnet clearly not learn the identification strong enough (mainly learn pose)
- 200+ steps with lora rank 16+ and lr 1e-5+ (several test conduct and also I check this: [Training Stable Diffusion with Dreambooth using Diffusers](#))
- Train text encoder: this is significant part, as we can see without it, the model weakly learn the subject (compare image 3 vs image 1 + 2 last slide)

A hard pose



2 prompts:

a photo of sks person

a photo of sks person in garden

Hard pose test

```
--num_class_images 100 \  
--train_batch_size 4 \  
--max_train_steps 800 \  
--learning_rate 5e-6 \  
--lora_rank 16
```

Cannot rendered face well

Kind of work with hard pose on base
prompt but fail completely on hard
prommpt

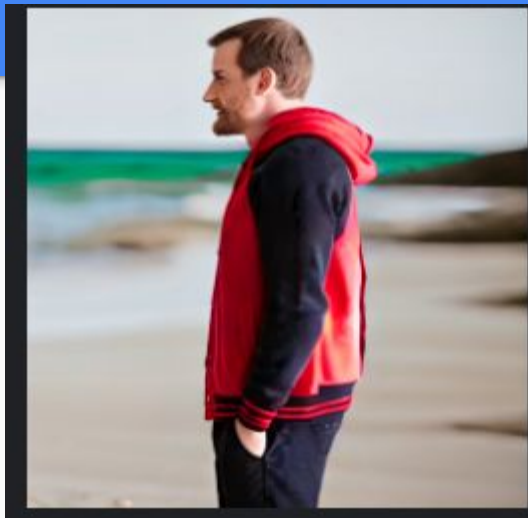


Hard pose test

```
--num_class_images 50 \  
--train_batch_size 4 \  
--max_train_steps 600 \  
--learning_rate 5e-5 \  
--lora_rank 64 \
```

Kind of work with hard pose on base
prompt but show better on hard
prompt

Better than before (larger lora and
lr rank improve training!)

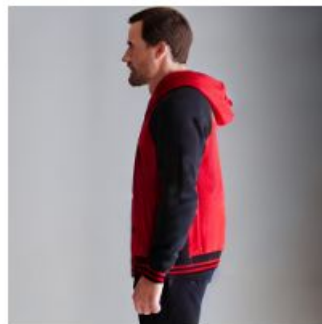
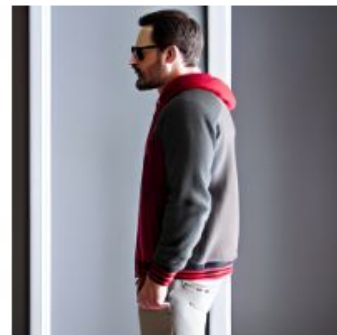
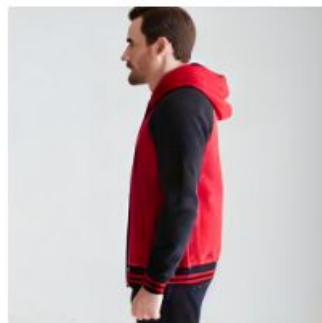


“a sks person with sunglass”



Overfit
(600 steps)

Underfit
(200 steps)



Overfit
(600 steps)

Underfit
(200 steps)

Conclusion for testing on human

We can see its really tricky:

- **Hard to balance** between overfit and underfit (a sks person with sun glass show this clearly as either the image ensemble training one or wrong outfit for subject (later, I know how to fix this through another paper :), but right now completely confused)
- Test on hard pose also tricky, as either under training wont work, or overfit will cause image fidelity (degrade in face generation)
- 200-600 (or on avg - 400) seem to be the right steps, higher lr + lowser lora or lower lr + higher lora all work pretty well (32/1e-4 or 64/1e-5 work most of the time)

Conclusion for testing on human

- Why plain dreambooth not work that well on pose conditioning T2I? My best assumption is that for each pose, we can have like 3-4 subject images that still satisfy the pose conditioning, but we limit the problem to 1 image per pose -> make it so much harder (the image just learn by heart paste the subject into the pose)
- Some future work I will test: for each pose, have about 2-3 subject images (seem hard to obtain) - generate prior image better (currently its quite of weak, which explain why its not that diverse after 600 steps)
- Some idea for generate prior, instead of off-policy generate and use prior loss, we could try on-policy generate (lets say after 100 steps, use the current model to generate new prior images?)

Conclusion for testing on human

- Evaluate criteria: follow prompt, understand the subject (human) and have the input pose
- Following input pose worked well
- Following prompt and understand subject would be a trade-off

Third task: Extend Dreambooth on humanoid

Problem Format:

Given a pretrained diffusion model

3-5 images of Unitree

A prompt describe the image want to be generated

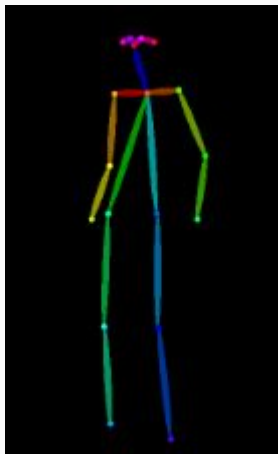
A pose

-> Finetuned so that the model can learn - follow prompt, understand the subject (human) and have the input pose

Controlnet + Dreambooth + humanoid

Data: collect from
<https://www.unitree.com/g1>

Pose: extracted from dataset use:
l1lyasviel/control_v11p_sd15_openpose
(pose extract model)!!!



First diff with human dataset

Openpose cannot extract all dataset i
have (I have 6, only 3 is extractable)!!!

So we use just 3 images for fine tuning

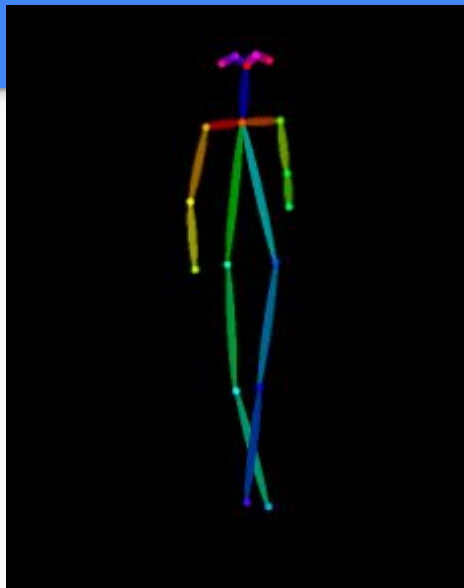
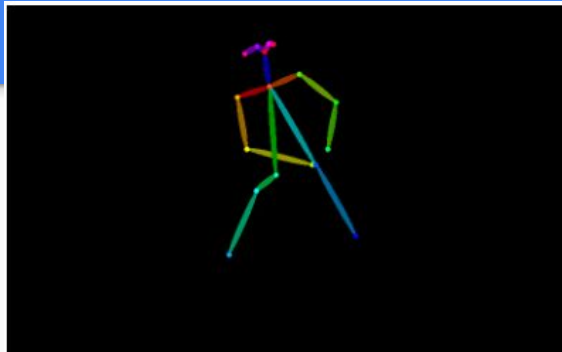
Second diff with human dataset

Pretrained ControlNet + “humanoid robot”
-> not really diverse generated images
(compare with human, we can see clearly
the **bias** in controlnet)

-> can affect the prior generation



Test case



Old pose with “a sks humanoid robot in an stadium”

-> check overfit - underfit

TEST

```
--num_class_images 20 \
```

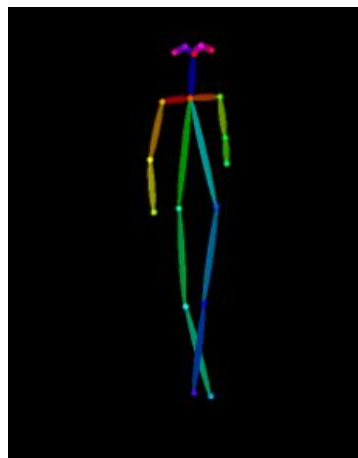
```
--train_batch_size 4 \
```

```
--max_train_steps 200 \
```

```
--learning_rate 1e-5 \
```

```
--lora_rank 64 \
```

Overfit + fail on hard pose

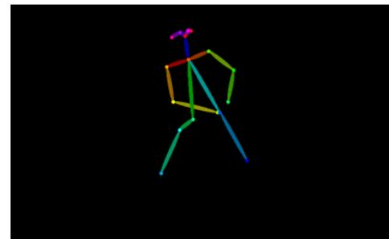
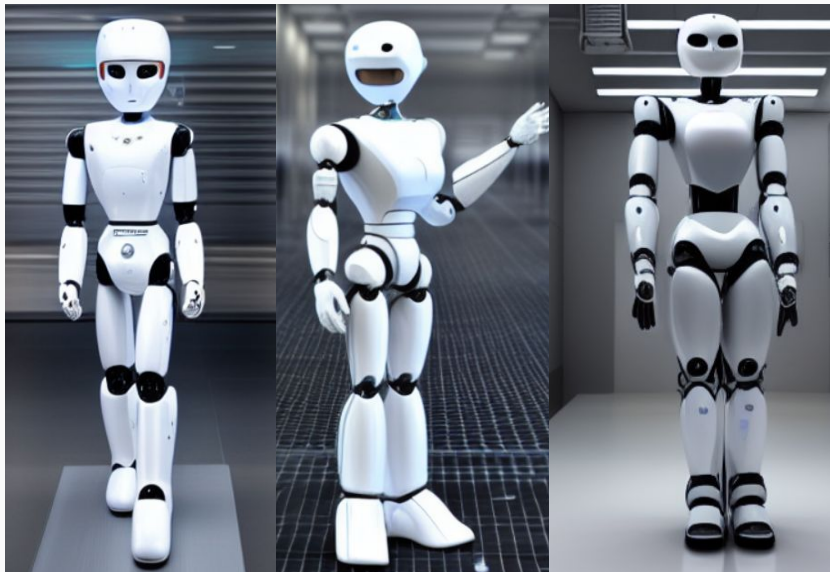


Third diff with human dataset

The hard human pose seem impossible for the controlnet -> proved by attempts to use pretrained model on this pose

The hard robot pose is still quite of possible

So I remove the human pose, keep the robot pose for test



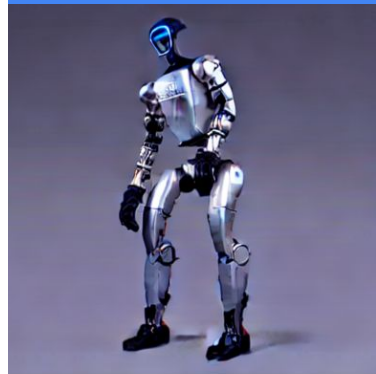
Follow pose + prompt but not learn Subject



Underfit
Not
Learned
The
Subject
well



Learn subject but not follow pose + prompt



Overfit

Take note
This is
Still
2 test
Like
above

TASK 4: Read more papers and Implement

From here, I give up the hyperparameter tuning approach and dive deeper into related papers to find if any can help

There are 2 main papers that significantly improve the work:

- <https://arxiv.org/pdf/2212.04488v2> (Custom Diffusion)
- <https://arxiv.org/pdf/2311.12052> (MagicPose)

CUSTOM DIFFUSION 1

`Train only K and V only in cross attention`

`(for my problem, I applied for K V training only in Lora)`

`Resize augmentation (for both images and prompts)`

- + **Random Resizing:** Input images are randomly resized between **0.4x** and **1.4x**.
- + **Prompt Adaptation:** The text prompt is dynamically updated to match the resize scale:
 - If resized small (<0.6x): Append **"very small"**, **"far away"**.
 - If resized large (>1.0x): Append **"zoomed in"**, **"close up"**.

Apply 2 key ideas from Custom Diffusion

COMPARE

First row:

hard pose

Second row:

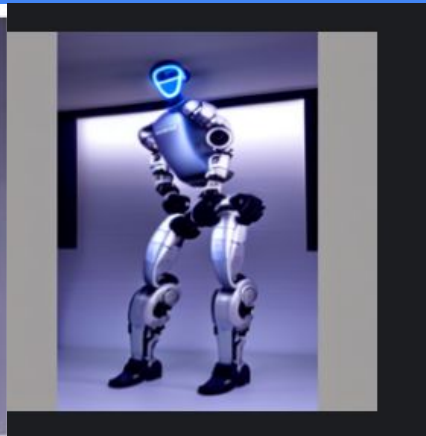
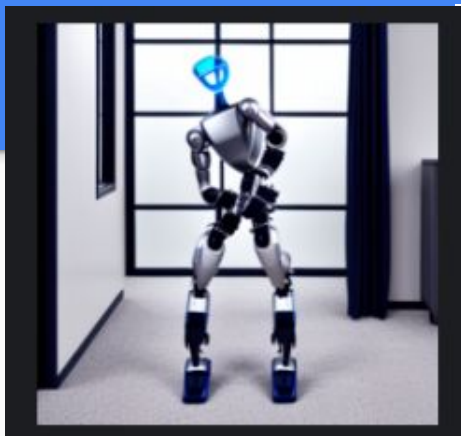
Stadium

The last row

is applied

Custom Diffusion

Show some balance!



Some more test on 2 key idea above on hard pose



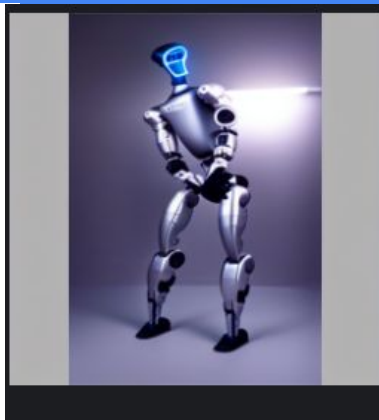
```
--num_class_images 20 \  
--train_batch_size 4 \  
--max_train_steps 100 \  
--learning_rate 1e-4 \  
--lora_rank 16 \  
--train_text_encoder \  

```



```
--num_class_images 20 \  
--train_batch_size 4 \  
--max_train_steps 200 \  
--learning_rate 1e-4 \  
--lora_rank 16 \  
--train_text_encoder \  

```



```
--num_class_images 100 \  
--train_batch_size 4 \  
--max_train_steps 200 \  
--learning_rate 1e-4 \  
--lora_rank 16 \  
--train_text_encoder \  

```



```
--num_class_images 100 \  
--prior_loss_weight 2.0 \  
--train_batch_size 4 \  
--max_train_steps 200 \  
--learning_rate 1e-4 \  
--lora_rank 32 \  
--train_text_encoder \  

```

CUSTOMER DIFFUSION 2

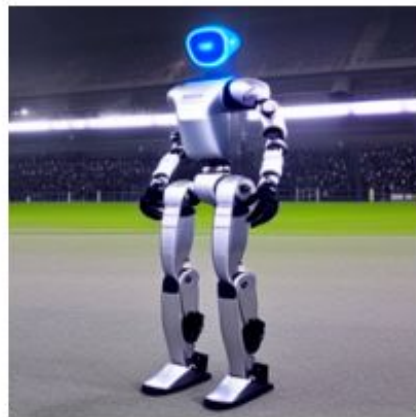
We still need a lot of work to find the best config for balance.

However, the key idea of reduce the amount of tuning parameter into K and V in cross attention only help a lots in keeping the pose

Here, I applied another key idea from Customer Diffusion:

- Instead of training full text encoder, we will train [V] token only! TRAIN TOKEN SKS ONL

Some test on 3rd key idea above



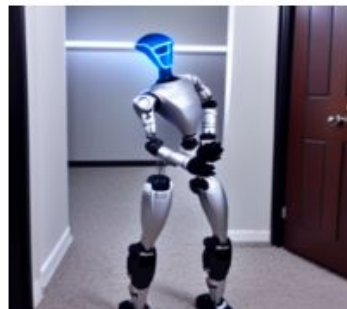
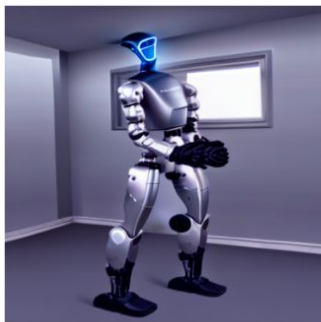
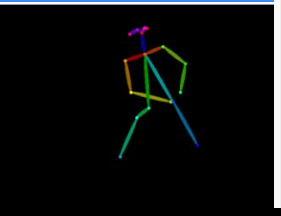
```
--num_class_images 100 \  
--train_batch_size 4 \  
--prior_loss_weight 2.0 \  
--max_train_steps 200 \  
--learning_rate 1e-4 \  
--lora_rank 32 \  
--train_token_embedding \  
--placeholder_token sks \  

```

```
--num_class_images 100 \  
--train_batch_size 4 \  
--max_train_steps 200 \  
--learning_rate 1e-4 \  
--lora_rank 32 \  
--train_token_embedding \  
--placeholder_token sks \  

```

Compare with underfitting



- The 1st row is applied text encoder training while the 2nd row is applied sks embedding training
- Overall, both underfitting but I will say the 2nd row keep the pose better when compare with hard pose, as the 1st row show some kind of catastrophic forgetting for posing when the robot now try to turn their face into the camera (note: all 3 training images turn their face into camera!)

Conclusion

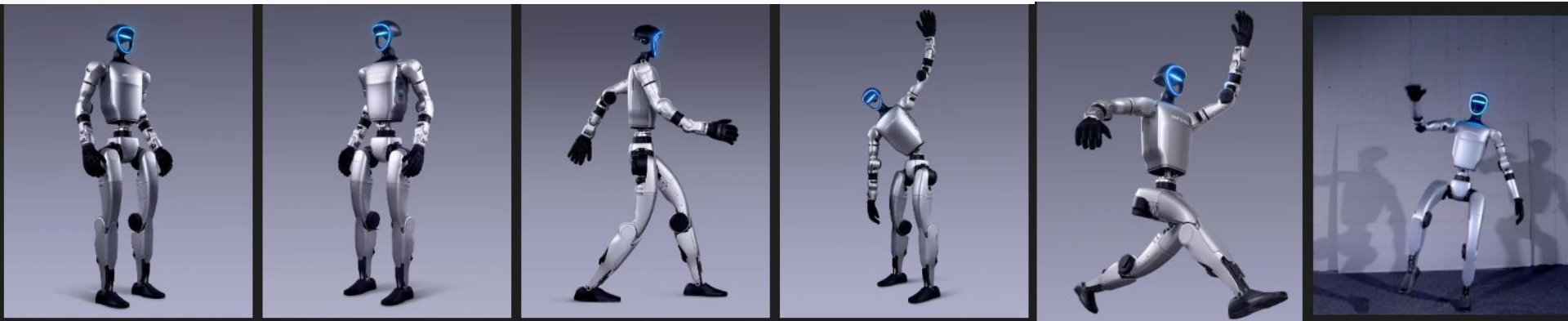
- With the third key idea, I now realize that training the entire text encoder is one of the reasons the model suffers from catastrophic forgetting.
- So, although applied the 3rd key idea show underfitting problem (Cannot learn the subject yet). I still use it for further experiments

MagicPose

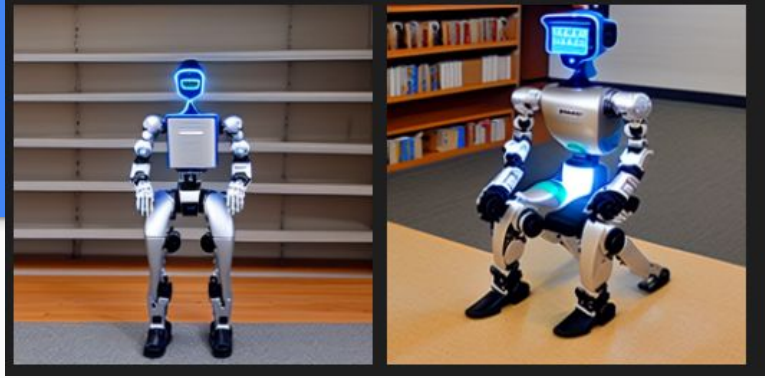
- The key idea I use from this paper is that splitting the training into 2 stages:
 - Appearance Control Pretraining: disable pose controlnet, train text encoder
 - Appearance-Disentangled Pose Control: disable text encoder, train pose

DATA STAGE 1

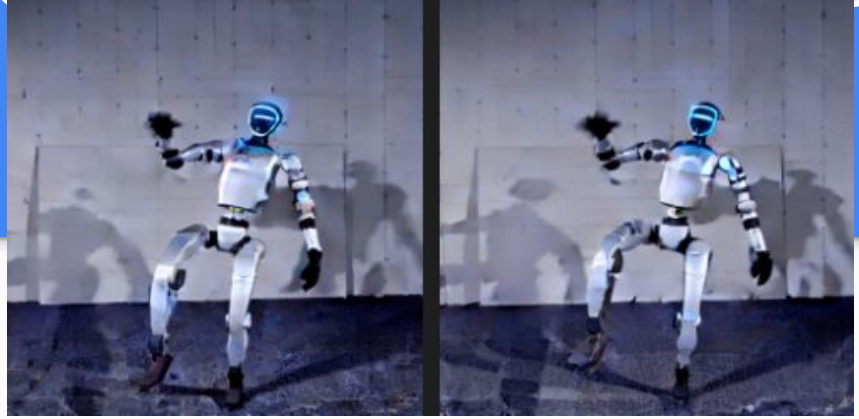
For stage 1, I can use up to 6 images below for training (A reminder that previous training, I still use 3 images only as I cannot extract the pose for all 6)



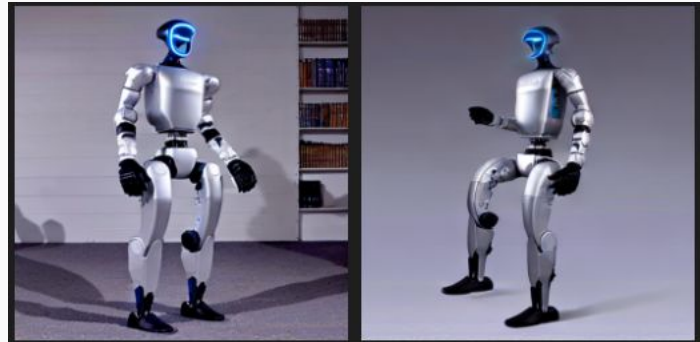
“A sks robot reading book” - sks embedding train



Training step: 200



Training step: 800



Training step: 400

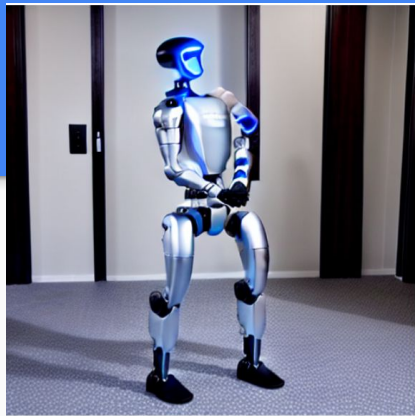
Conclusion

- Obviously, 800 steps show clear overfitting, and 400 steps still have bookselves in first images, but the 2nd image is irrelevant
- For the 200 steps, we see that the robot is in library, so its the one following the prompt the most - although its not learn the robot appearance + logic pose that well yet!
- But that for stage 2!

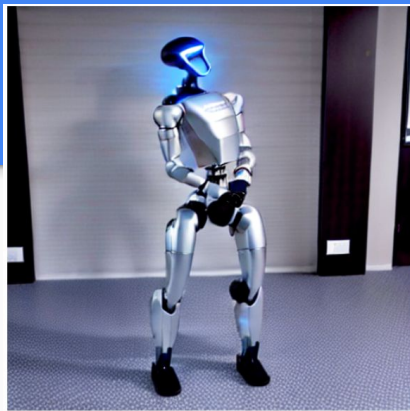
DATA STAGE 2

Same as other pose training

Some test on 2nd stage



```
--num_class_images 10 \  
--train_batch_size 4 \  
--max_train_steps 20 \  
--learning_rate 1e-5 \  
--controlnet_learning_rate  
--lora_rank 64 \  
--controlnet_lora  
(200 steps from stage 1)
```



```
--num_class_images 20 \  
--train_batch_size 4 \  
--max_train_steps 100 \  
--learning_rate 1e-5 \  
--no_train_controlnet \  
--lora_rank 32  
(200 steps from stage 1)
```



```
--num_class_images 20 \  
--train_batch_size 4 \  
--max_train_steps 40 \  
--learning_rate 1e-5 \  
--no_train_controlnet  
--lora_rank 32  
(400 steps from stage 1)
```

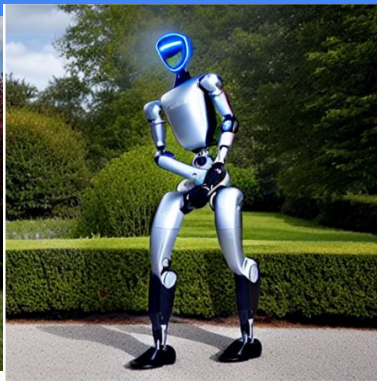


```
--num_class_images 20 \  
--train_batch_size 4 \  
--max_train_steps 100 \  
--learning_rate 1e-5 \  
--controlnet_learning_rate  
--lora_rank 64 \  
--controlnet_lora  
(400 steps from stage 1)
```

Conclusion

- We see that slightly training on pose lead to some degrade in robot appearance
- Use 400 steps from stage 1 lead to overfit

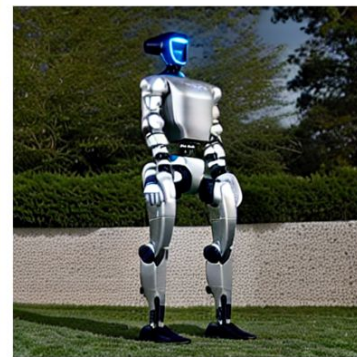
More test on hard pose with “sks in garden”



200 steps from stage 1



400 steps from stage 1



800 steps from stage 1

Final Conclusion

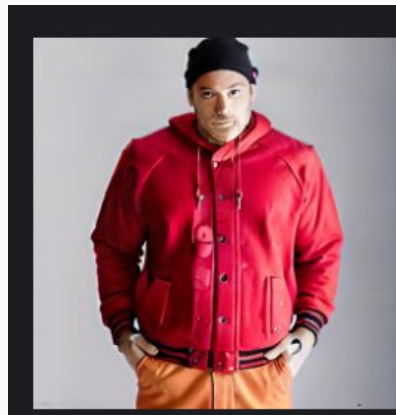
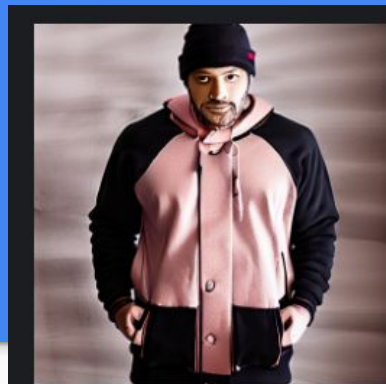
- In conclusion, I still cannot make a model that can fully make the perfect image on the hard pose
- As I show in the whole progress above, the initial controlnet already show some struggle with humanoid robot and hard pose, also with the limited data + pose, but I still have a lot of thing to try and learn, so I hope that my attempts in step by step exploring the problems/handling the difficulty/intensively try-and-error could show my whole dedicated work on the last 2 weeks
- Also, I have a small ablation study section that show some more testing and key observation.

ABLATION STUDY

Some other papers I read

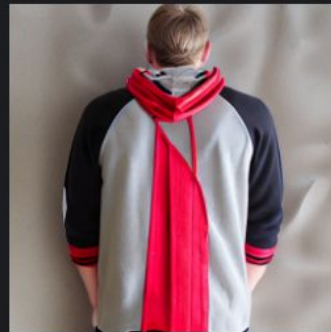
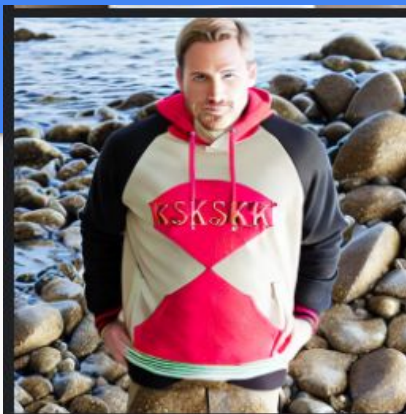
- Textual Inversion: [\[2208.01618\] An Image is Worth One Word: Personalizing Text-to-Image Generation using Textual Inversion](#)
- CTIG-DM: [\[2306.10804\] Conditional Text Image Generation with Diffusion Models](#)
- T2I-Adapter: [\[2302.08453\] T2I-Adapter: Learning Adapters to Dig out More Controllable Ability for Text-to-Image Diffusion Models](#)
- GLIDE: [\[2112.10741\] GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models](#)
- DreamPose: [\[2304.06025\] DreamPose: Fashion Image-to-Video Synthesis via Stable Diffusion](#)
- DisenBooth: [\[2305.03374\] DisenBooth: Identity-Preserving Disentangled Tuning for Subject-Driven Text-to-Image Generation](#)
- PIDM: [\[2211.12500\] Person Image Synthesis via Denoising Diffusion Model](#)
- PoCoLD: [Controllable Person Image Synthesis with Pose-Constrained Latent Diffusion](#)
- HumanSD: [\[2304.04269\] HumanSD: A Native Skeleton-Guided Diffusion Model for Human Image Generation](#)

Underfit by freezing text encoder



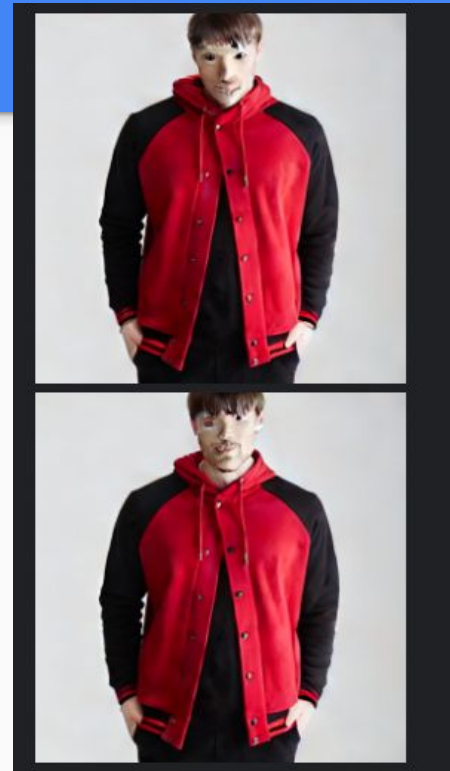
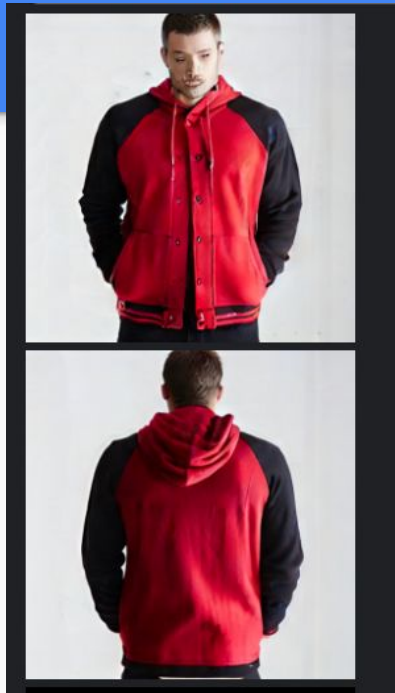
Underfit by weak params

All is trained on 200 steps

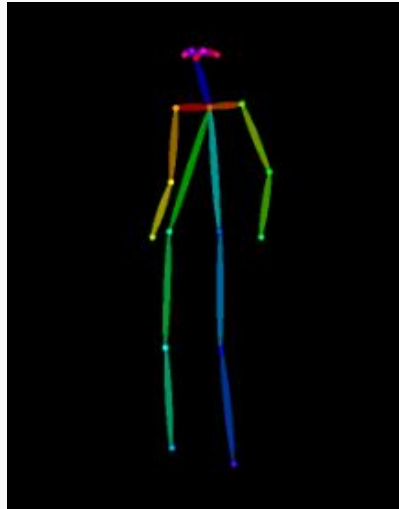


Degraded face for over 800 steps

All is trained on 800 steps



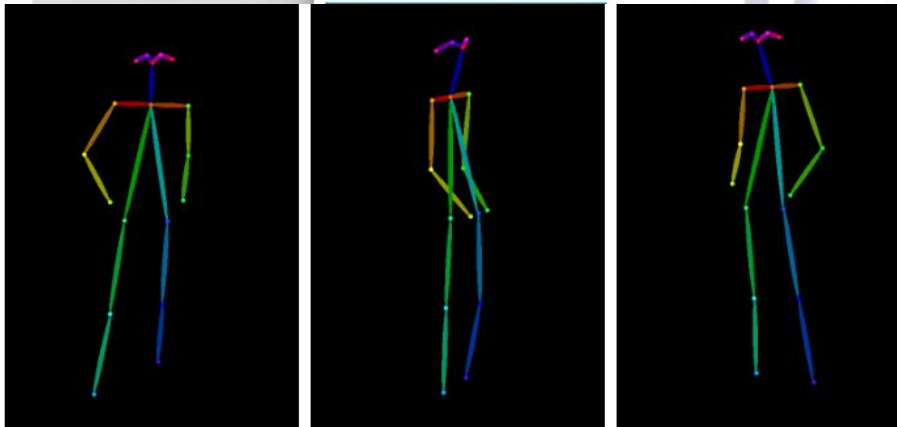
Test stage 1 on pretrained pose conditioning (without stage 2)



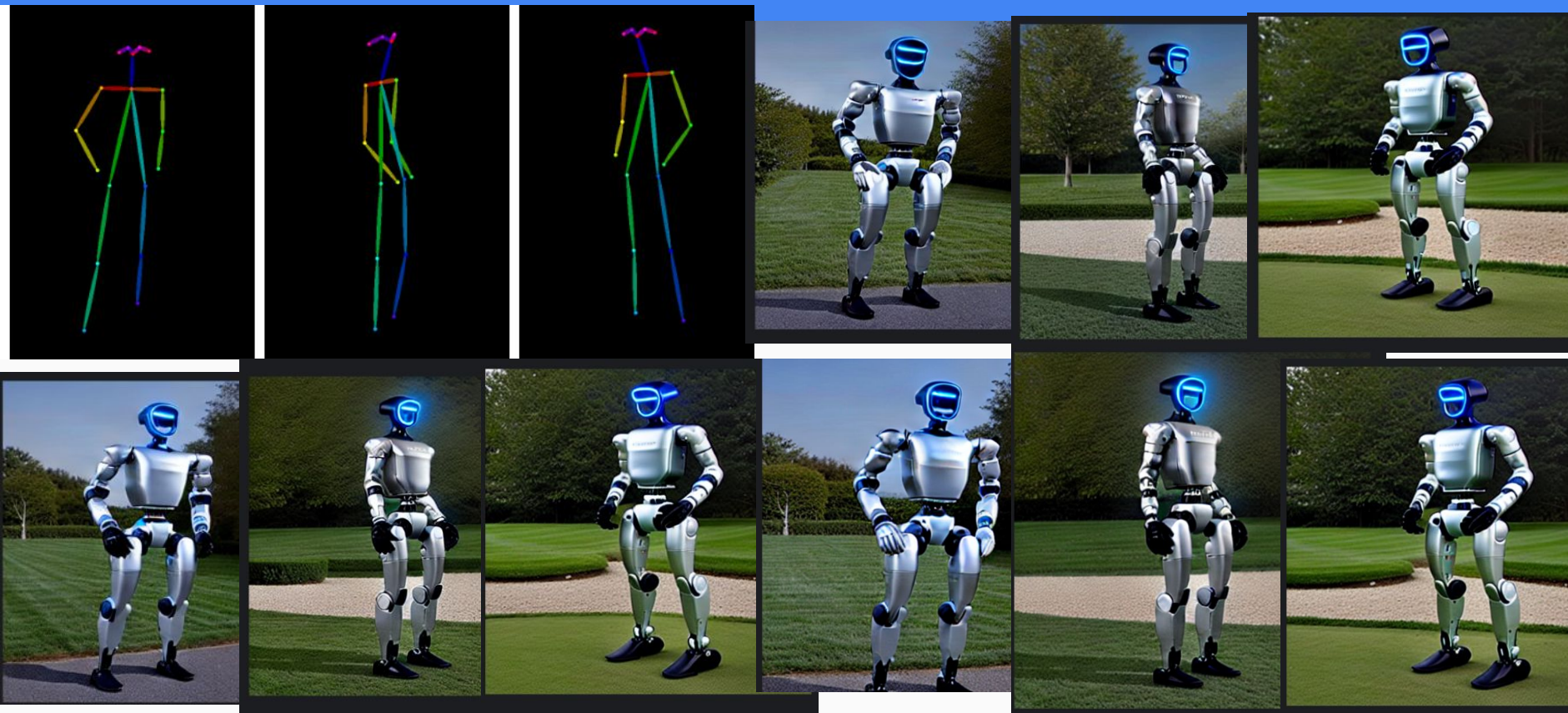
Some more (easier) pose to test on stage 2



a sks humanoid robot in garden



Test on(easier) pose to test on stage 2 (1)



Test on(easier) pose to test on stage 2 (2)

